## Nomenclature

| | | |
|---|---|---|
| $M$ | = | Inertia matrix, combining rigid-body and added mass effects |
| $C(\nu)$ | = | Coriolis and centripetal matrix, combining rigid-body and added mass effects |
| $D(\nu)$ | = | Vector of damping forces |
| $g(\eta)$ | = | Vector of restoring forces |
| $J(\nu)$ | = | Coordinate transformation matrix |
| CO | = | origin of the body frame |
| $X_u, Y_v, Z_w$ | = | Linear damping coefficients for translation in water |
| $K_p, M_q, N_r$ | = | Linear damping coefficients for rotation in water |
| $X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}$ | = | Increased inertia about x, y, z axis due to translation in water |
| $K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}$ | = | Increased inertia about x, y, z axis due to rotation in water |
| $X_{u\|u\|}, Y_{v\|v\|}, Z_{w\|w\|}$ | = | Quadratic damping coefficients for translation in water |
| $K_{p\|p}, M_{q\|q}, N_{r\|r}$ | = | Quadratic damping coefficients for rotation in water |
| $W, B$ | = | weight and buoyant force |
| $m$ | = | mass of ROV |

## 1. Foseen's notation

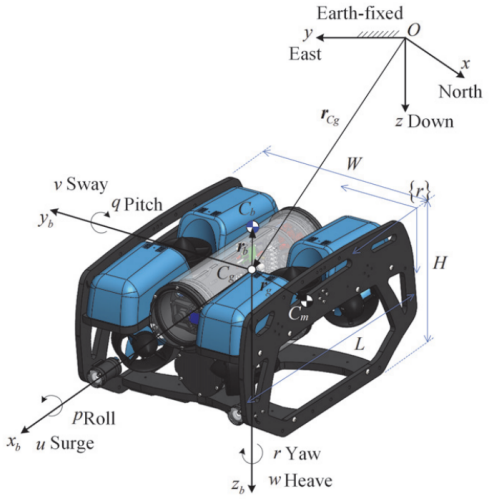



**Fig. 1 BlueROV2 reference frame [1]**

Generalized coordinates:

$$\eta = [x, y, z, \phi, \theta, \psi]^T$$

Generalized velocity:

$$\nu = [u, v, w, p, q, r]^T$$

Generalized force:

$$\tau = [X, Y, Z, K, M, N]^T$$

** 6 thruster configuration has no pitch control **

## 2. Kinematics

$$\Theta = [\phi, \theta, \psi]^T$$

Transformation matrix is:

$$J(\eta) = \begin{bmatrix} R(\Theta) & 0_{3\times3} \\ 0_{3\times3} & T(\Theta) \end{bmatrix}$$

where

$$R(\Theta) = \begin{bmatrix} c\psi \cdot c\theta & -s\psi \cdot c\phi + c\psi \cdot s\theta \cdot s\phi & s\psi \cdot s\phi + c\psi \cdot c\phi \cdot s\theta \\ s\psi \cdot c\theta & c\psi \cdot c\phi + s\phi \cdot s\theta \cdot s\psi & -c\psi \cdot s\phi + s\theta \cdot s\psi \cdot c\phi \\ -s\theta & c\theta \cdot s\phi & c\theta \cdot c\phi \end{bmatrix}$$

$$T(\Theta) = \begin{bmatrix} 1 & s\phi \cdot t\theta & c\phi \cdot t\theta \\ 0 & c\theta & -s\theta \\ 0 & \frac{s\phi}{c\theta} & \frac{c\phi}{c\theta} \end{bmatrix}$$

with $s\cdot = \sin(\cdot)$, $c\cdot = \cos(\cdot)$, $t\cdot = \tan(\cdot)$.

## 3. Assumptions

Assumption for now:

1) Assume CB is at CO $(x_b, y_b, z_b) = (0, 0, 0)$ and CG is directly below CO $(x_g, y_g, z_g) = (0, 0, z_g)$.
2) Assume symmetric $I_{xy} = I_{xz} = I_{yz} = 0$.
3) Assume neutrally buoyant $mg = \rho g V$.
4) Underwater currents are constant.

## 4. Kinetics

Let $(x_g, y_g, z_g)$ be the center of gravity (CG) and $(x_b, y_b, z_b)$ be the center of buoyancy (CB) measured from the center of origin (CO).

Rigid body inertia matrix:

$$M_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & I_{xy} & I_{xz} \\ mz_g & 0 & -mx_g & I_{yx} & I_y & I_{yz} \\ -my_g & mx_g & 0 & I_{zx} & I_{zy} & I_z \end{bmatrix}$$

Apply assumptions, it becomes:

$$M_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & 0 \\ 0 & m & 0 & -mz_g & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & -mz_g & 0 & I_x & 0 & 0 \\ mz_g & 0 & 0 & 0 & I_y & 0 \\ 0 & 0 & 0 & 0 & 0 & I_z \end{bmatrix}$$

Added-mass inertia matrix (assume symmetric, off-diagonal terms are often hard to determine):

$$M_A = -\operatorname{diag}\{X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}\}$$

$$M = M_{RB} + M_A$$

Rigid body coriolis matrix:

$$C_{RB} = \begin{bmatrix} 0 & 0 & 0 & m(y_G q + z_G r) & -m(x_G q - w) & -m(x_G r + v) \\ 0 & 0 & 0 & -m(x_G q + w) & m(z_G r + x_G p) & -m(y_G r - u) \\ 0 & 0 & 0 & -m(x_G r - v) & -m(z_G q + u) & m(x_G p + y_G q) \\ -m(y_G q + z_G r) & m(y_G p + w) & m(z_G p - v) & 0 & I_z q & -I_y p \\ m(x_G q - w) & -m(z_G r + x_G p) & -m(x_G p + y_G q) & -I_z q & 0 & I_x p \\ m(x_G r + v) & m(y_G r - u) & -m(x_G p + y_G q) & I_y q & -I_x p & 0 \end{bmatrix}$$

Apply assumptions, it becomes:

$$C_{RB} = \begin{bmatrix} 0 & 0 & 0 & mz_G r & mw & -mv \\ 0 & 0 & 0 & -mw & mz_G r & mu \\ 0 & 0 & 0 & mv & -m(z_G q + u) & 0 \\ -mz_G r & mw & m(z_G p - v) & 0 & I_z q & -I_y p \\ -mw & -mz_G r & 0 & -I_z q & 0 & I_x p \\ mv & -mu & 0 & I_y q & -I_x p & 0 \end{bmatrix}$$

Added mass Coriolis force matrix

$$C_A = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}} w & Y_{\dot{v}} v \\ 0 & 0 & 0 & Z_{\dot{w}} w & 0 & -X_{\dot{u}} u \\ 0 & 0 & 0 & -Y_{\dot{v}} v & X_{\dot{u}} u & 0 \\ 0 & -Z_{\dot{w}} w & Y_{\dot{v}} v & 0 & -N_{\dot{r}} r & M_{\dot{q}} q \\ Z_{\dot{w}} w & 0 & -X_{\dot{u}} u & N_{\dot{r}} r & 0 & -K_{\dot{p}} p \\ -Y_{\dot{v}} v & X_{\dot{u}} u & 0 & -M_{\dot{q}} q & K_{\dot{p}} p & 0 \end{bmatrix}$$

$$C(\nu) = C_{RB} + C_A$$

Damping matrix (addition of nonlinear and linear):

$$D = -\text{diag}\{X_{u|u|}|u| + X_u, Y_{v|v|}|v| + Y_v, Z_{w|w|}|w| + Z_w, K_{p|p|}|p| + K_p, M_{q|q|}|q| + M_q, N_{r|r|}|r| + N_r\}$$

Restoring forces vector:

$$g(\nu) = -\begin{bmatrix} (W-B)s\theta \\ -(W-B)c\theta \cdot s\phi \\ -(W-B)c\theta \cdot c\phi \\ (z_G W - z_B B)c\theta \cdot s\phi \\ (z_G W - z_B B)s\theta + (x_H W - x_B B)c\theta \cdot c\phi \\ -(x_G W - x_B B)c\theta \cdot s\phi \end{bmatrix}$$

Apply assumptions, it becomes:

$$g(\nu) = -\begin{bmatrix} 0 \\ 0 \\ 0 \\ z_G W c\theta \cdot s\phi \\ z_G W s\theta \\ 0 \end{bmatrix}$$

## 5. EOM

$$\dot{\eta} = J(\eta)\nu \tag{1}$$

$$M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau \tag{2}$$

Map thrust to PWN by curve-fitting (assume 18V):

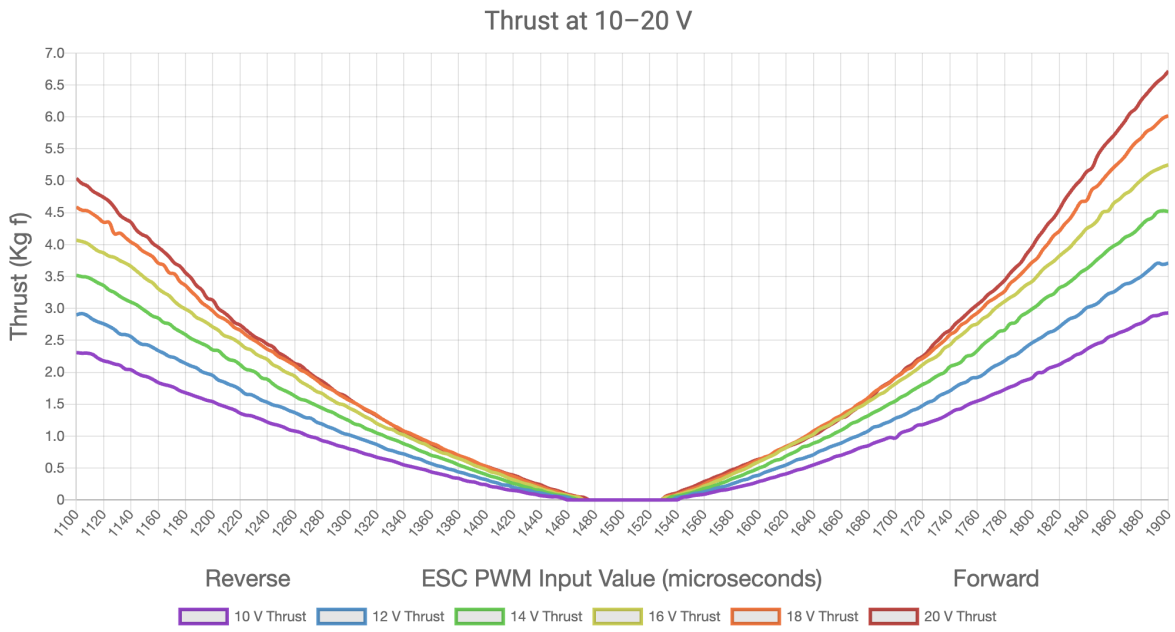



**Fig. 2 T200 Thrust vs. PWN from BlueRobotics**

```
def force_to_pwm(F):
    F_dead = 0.1
    if abs(F) <= F_dead:
        return 1500
    elif F > 0:
        F = F / 9.81 # convert to kg f
        return -5.88777075*F**2 + 92.44646297*F + 1541.40847528
    else:
        F = F / 9.81 # convert to kg f
        return 8.3970206*F**2 + 113.7834968*F + 1460.38642332
```

** changed $F_{dead}$ to 0.3 after testing

```
def force_to_pwm(F):
    if abs(F) < 0.3:
        return 1500
    F = F / 9.81
    if F > 0:
        pwm = -5.887*F**2 + 92.45*F + 1541.4
    else:
        pwm = 8.397*F**2 + 113.78*F + 1460.4
    return int(np.clip(pwm, 1100, 1900))
```

## References

[1] Einarsson, E. M., and Lipenitis, A., "Model Predictive Control for the BlueROV2 Theory and Implementation," *Aalborg Universitet*, 2020. URL https://projekter.aau.dk/projekter/files/387609647/MPC_control_for_the_BlueROV2_Theory_and_Implementation.pdf.

[2] von Benzon, M., Sørensen, F. F., Uth, E., Jouffroy, J., Liniger, J., and Pedersen, S., "An Open-Source Benchmark Simulator: Control of a BlueROV2 Underwater Robot," *Journal of Marine Science and Engineering 2022, Vol. 10,*, Vol. 10, 2022. https://doi.org/10.3390/JMSE10121898, URL https://www.mdpi.com/2077-1312/10/12/1898.

[3] Wu, C.-J., "6-DoF Modelling and Control of a Remotely Operated Vehicle," 2018.

## Appendix

| Symbol | Value | Unit |
|---|---|---|
| $m$ | 11.4 | $kg$ |
| $I_x$ | 0.21 | $kgm^2$ |
| $I_y$ | 0.245 | $kgm^2$ |
| $I_z$ | 0.245 | $kgm^2$ |
| $g$ | 9.82 | $kg/m^2$ |
| $z_g$ | 0.02 | $m$ |
| $X_u$ | 13.7 | $Ns/m$ |
| $X_{u\|u\|}$ | 141.0 | $Ns^2/m^2$ |
| $Y_v$ | 0 | $Ns/m$ |
| $Y_{v\|v\|}$ | 217.0 | $Ns^2/m^2$ |
| $Z_w$ | 33.0 | $Ns/m$ |
| $Z_{w\|w\|}$ | 190.0 | $Ns^2/m^2$ |
| $K_p$ | 0 | $Ns/m$ |
| $K_{p\|p\|}$ | 1.19 | $Ns^2/m^2$ |
| $M_q$ | 0.8 | $Ns$ |
| $M_{q\|q\|}$ | 0.47 | $Ns^2$ |
| $N_r$ | 0 | $Ns$ |
| $N_{r\|r\|}$ | 1.5 | $Ns^2$ |
| $X_{\dot{u}}$ | 6.36 | $kg$ |
| $Y_{\dot{v}}$ | 7.12 | $kg$ |
| $Z_{\dot{w}}$ | 18.68 | $kg$ |
| $K_{\dot{p}}$ | 0.189 | $kgm^2$ |
| $M_{\dot{q}}$ | 0.135 | $kgm^2$ |
| $N_{\dot{r}}$ | 0.222 | $kgm^2$ |

**Table 2 Estimated Coefficients [1] [2]**

Also, see [3]: uses negative added mass, linear damping, and quadratic damping coefficients.